Topic Outline

2.1 Extraction

2.1.1 Definition and the need of Extraction

2.1.2 Preparation of plant material for solvent extraction

2.1.3 Choice of suitable solvents

2.1.4 Methods of extraction

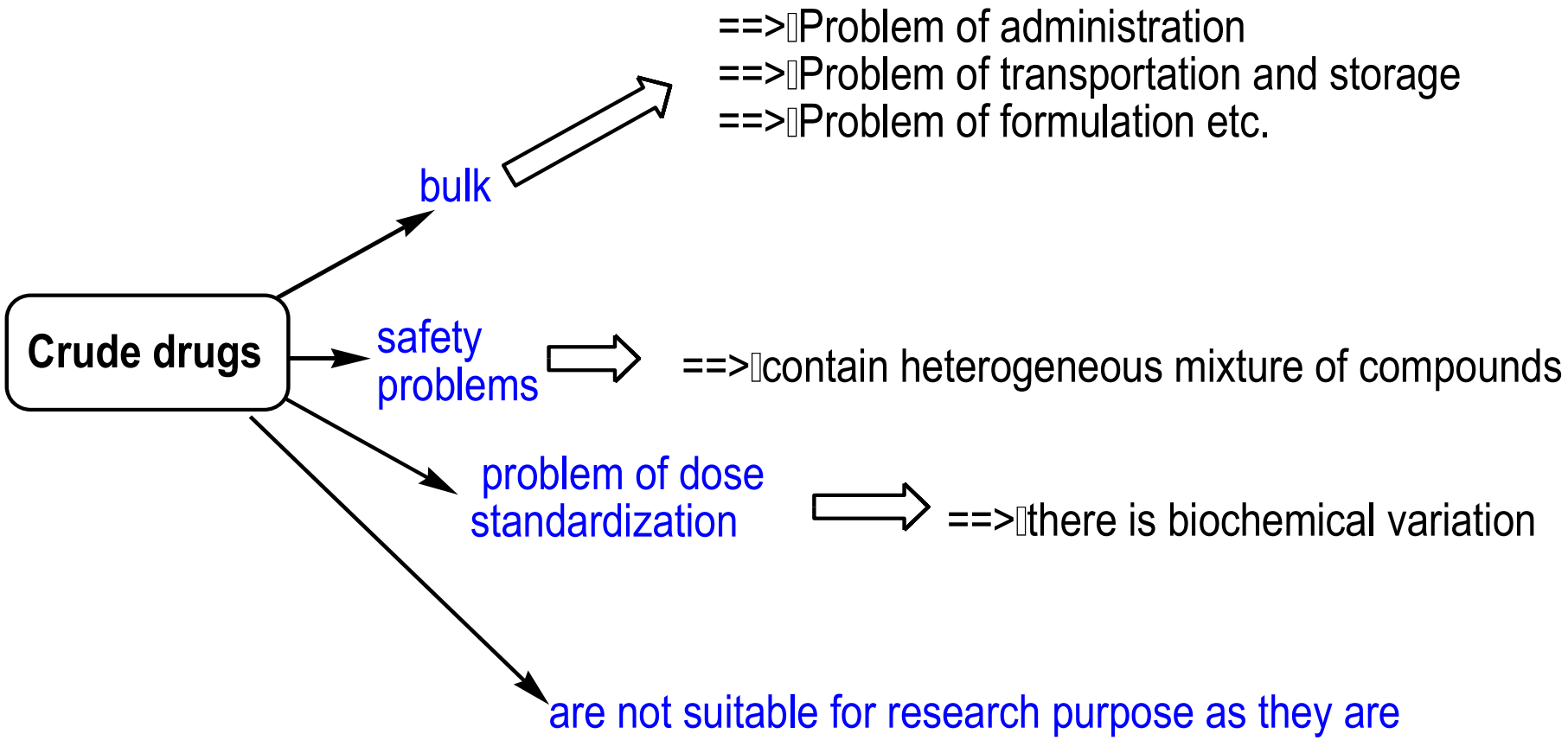
- Maceration/Percolation, Refluxing, Soxhlet or continuous extraction, and Gradient extraction
- **Distillation**: Hydro distillation, Steam distillation, Water and stem distillation

Extraction

- Introduction
  - In order to study plant constituents they must be extracted from the plant.
  - Crude drugs are vegetable or animal drugs that have undergone no other processes than collection and drying. But they may be powdered.
  - Crude drugs of natural origin can be categorized as :-
    - organized (cellular) drugs and
      - ✓ Include digitalis leaves, cinchona bark, sandal wood, rauwolfia roots, Nux-vomica seed
    - Unorganized (acellular) drugs.
      - ✓ Include plant exudates, plant latex, euphoria latex and extracts of seeds

A. The need for extraction

- Crude drugs are not usually directly used as therapeutic agents because:-



The Need for extraction

- Therefore,
  - It is required that the active principles are separated from crude drugs
  - Various methods of separation may be employed
  - Extraction is a generally accepted method
- Thus,
  - In order to study plant constituents they must be extracted from the plant first.

The need for extraction cont.

- Moreover,
  - Previously crude drugs were mainly used in the form of alcoholic extracts
  - Nowadays, pure isolated active principles are used
  - This is not always possible because:
    - It may be difficult to separate due to more economic to use extracts
    - Some active principles are unstable when isolated
    - Active principles not known – activity thought from mixture
- Pharmacist needs basic knowledge of
  - the ways in which crude drugs can be extracted and tested for presence of active principles
  - quality assurance of crude drugs

Extraction

- It is the process of obtaining desired constituent(s) from crude drugs through the use of selected solvent(s) in which the desired constituents are soluble
- Generally, extraction is the process of obtaining the extractable components of plants or animal tissues treated with specific solvents whereby the medicinally active constituents are dissolved out, cell tissues and most of inactive or inert components remain undissolved.
- The plant material should be properly authenticated or identified before extraction.
- The process of drug extraction is a generally accepted method of obtaining the active principles.

- Crude drugs contain a number of constituents that may be soluble in a given solvent(s), and hence
  - The products of extraction from these crude drugs are called **extractives** or **crude extracts**.
  - Crude extracts contain varying number of constituents, depending upon the drug used and the condition of extraction.
- The solvent or solvent-mixtures used is referred to as **menstruum**, and
- The drug residue which is exhausted of active constituent is termed the **marc**.

- **Components of crude drugs**
  - **Extractable**
    - Low molecular wts.
    - Can be dissolved in common solvents
    - Can be intracellular or extracellular plant constituents
  - **Non-extractable**
    - Biological polymers and insoluble in common solvents and include
      - structural components like cellulose and structural proteins
      - Storage polysaccharides
      - Enzymes, nucleic acids etc.
- Ideally an extraction method should
  - preserve the plant constituents
  - be preferably fast
  - be efficient
  - be economical
  - be selective to the medicinally important plant components

- The inert substance
  - ➢ May interferes with the stability of
    - ⇒ either the active constituent or
    - ⇒ the final crude extract
  - ➢ May, in some way, make the preparation
    - ⇒ discolored or unpalatable
  - ➢ In such cases, steps would be taken
    - ⇒ either to prevent its extraction or
    - ⇒ to remove it from the extractive after its extraction.

Preparation of crude drugs for extraction

- Plant material is collected and separated
- Most crude drugs are **dried** soon after collection to
  - Reduce any enzymatic activity that could result in destruction of the active constituents;
  - Prevent the growth of bacteria, mold and yeast
  - Facilitate powdering, packaging, storage and transportation
  - Decreases deterioration and infestation.
  - Allow easy penetration of solvent
    - If un-dried there is water in the plant material ⇒ formation of emulsion with organic solvents ⇒ difficult to break and decease yield of extraction

Preparation of crude drugs for extraction

- Before a plant drug is dried, the active constituents are generally present within the cells in
  - aqueous solution or
  - In colloidal suspension.
- After drying, the active constitutes occur within the dried cells as
  - crystalline or
  - amorphous substances.
- **Note**:
  - drying should be done as quickly as possible
  - Drying conditions should be considered, eg
    - Drying temperature ≤ 40°C
    - Dry under the shade (avoid direct sunlight)

Preparation for extraction cont.

- The dried plant materials are generally powdered
- Powdering
  - ➢ break the majority of the cell walls and enhance solvent penetration
  - ➢ increase surface area

The menstruum

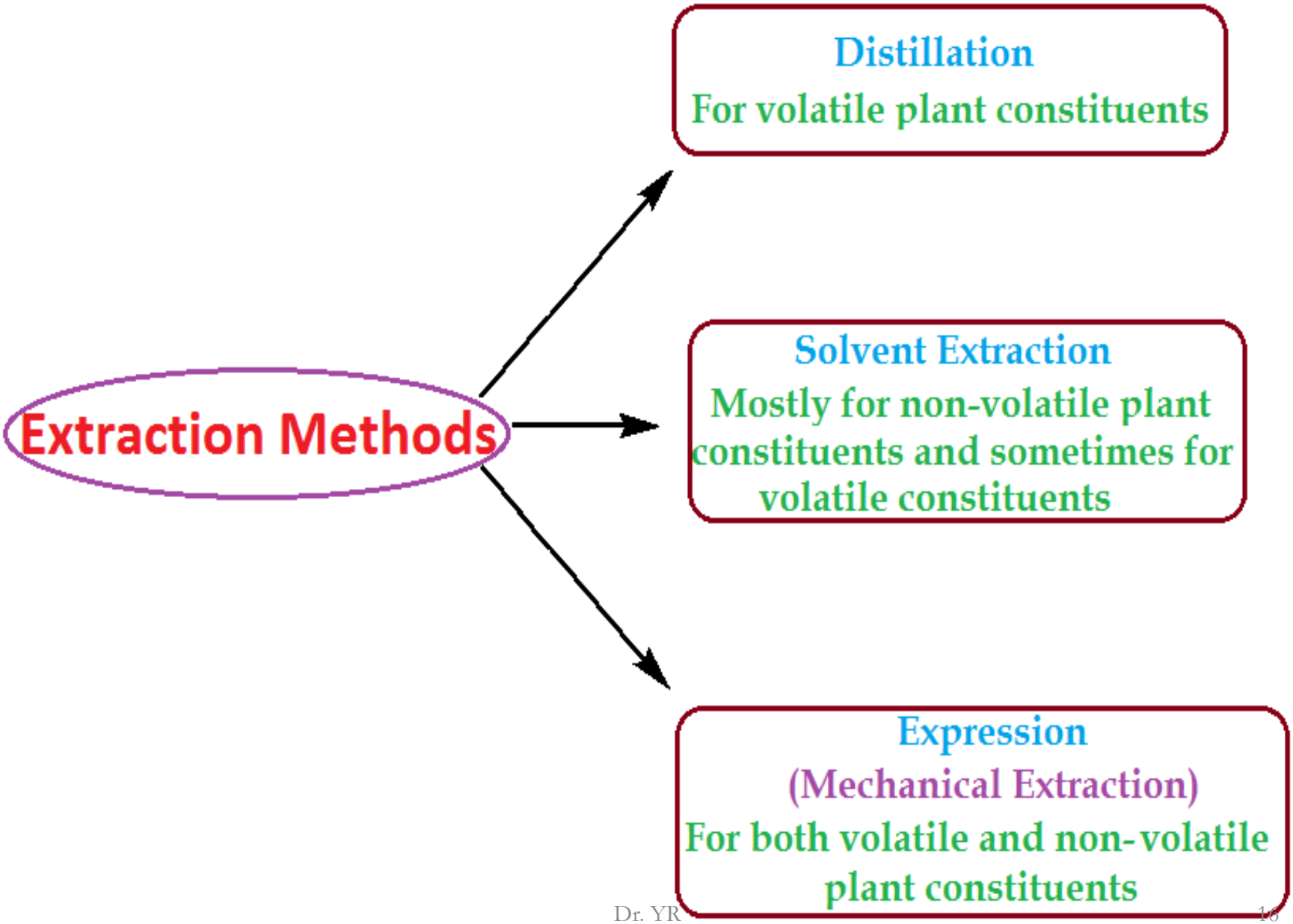
- Ideal properties of extraction solvent(s):-
  - It should
    - be readily available
    - be cheap
    - have good solvent action for many plant constituents
    - be non-toxic, non-irritant and nonflammable.
    - be environmentally friendly
    - be non-explosive
    - not have any destructive effect to the constituent to be extracted
    - penetrate the plant material and reach where the plant constituent is found

- The selection of the menstruum is based primarily upon the relative solubility of
  - the active constituents and
  - inactive and undesirable constituents in it.
- Alcohol is a general solvent for many plant constituents
- 70% alcohol is used as a universal solvent for extraction
- Water immiscible solvents are also widely used:-
  - Light petroleum, benzene, hexane for extraction of essential oils, fixed oils, and steroids
  - Ether and chloroform for alkaloids and quinones etc
- Extraction is effected mainly due to solubility and to some extent due to diffusion.
- The choice of solvent depends mainly on the solubility of the plant constituents of interest.
    - Remember that "like dissolves like".

Solvents used for active component extraction

Water	**Ethanol**	**Methanol**	**Chloroform**	**Ether**	**Acetone**
Anthocyanins	Tannins	Anthocyanins	Terpenoids	Alkaloids	Phenol
Starches	Polyphenols	Terpenoids	Flavonoids	Terpenoids	Flavonols
Tannins	Polyacetylenes	Saponins		Coumarins	
Saponins	Flavonol	Tannins		Fatty acids	
Terpenoids	Terpenoids	Xanthoxyllines			
Polypeptides	Sterols	Totarol			
Lectins	Alkaloids	Quassinoids			
		Lactones			
		Flavones			
		Phenones			
		Polyphenols			

Classification of Extraction methods



Solvent extraction

- Solvent extraction is a method of extraction where the plant material is extracted with a solvent, which could be
  - liquid,
  - semisolid or
  - Supercritical fluid (CO_2 or N_2).
- Thus, depending on the volatility of the solvent, extraction can be:
  - volatile solvent extraction
  - non-volatile solvent extraction and
  - Supercritical solvent extraction.

- **Solvent extraction methods (Extraction techniques)**
  - –The principal methods of drug extraction are:-
    - Maceration
    - Percolation
      - Cold
      - Hot continuous percolation process (Soxhlet)
    - Infusion
    - Digestion
    - Decoction
    - Refluxing
- The choice of the extraction procedure generally depends on:-
  - – the nature of the crude drug ;
  - – the nature of the components to be isolated
  - – purpose of extraction
  - – availability of solvent and apparatus etc

Maceration

- Term derived from the latin word
  - ✓"**macerare**" meaning to soak
  - ✓Maceration simply means to soak
- It is a process in which the properly powdered drug is allowed to soak in the menstruum untill
  - The cellular structure is softened and penetrated by the menstruum,
  - The soluble constituents are dissolved and
  - It is usually done at a temperature between 15ºC to 20ºC for a period of 3 days according to USP.

Maceration

Procedure:

- Plant material
  - Is mixed with (soaked in) the solvent in a closed container and
  - Allowed to stand for at least 3 days (72 hours), with occasional stirring.
  - The container or vessel is stoppered tightly
  - The contents are agitated or stirred occasionally over a period.
  - The extract is filtered and the filtrate is dried.

Maceration

- Repeated maceration may be more effective than single maceration because
  - an appreciable amount of active constituents may be left behind in the first pressing of marc which may be extracted out in the next maceration.
- It is established that maximum extraction is obtained in multiple maceration when the total quantity of menstruum is present during each maceration.
- Double maceration is carried out twice and the total volume of the menstruum to be used is divided into two parts.
- In triple maceration is the menstruum is divided into three parts.

Maceration

- **Advantage**:
  - Simple.
  - Thermolabile compounds can be extracted
- **Disadvantage**:
  - Consumes too much solvent
  - Saturation is attained quickly.
  - Time taking process with the possibly of hydrolysis of compounds
    - Solubility and diffusion are involved in the extraction process.
    - Usually done at room temperature.

Percolation

- The term percolation comes from the Latin words
  - per means through and
  - colate mean to strain.
- May be described generally as a process in which powdered drug is extracted by slow passage of a suitable solvent through a column of drug.
- The powdered drug is packed in a special extraction apparatus termed as percolator and the extractive is collected termed as percolate.

Percolation

Cold Percolation

- In this process the flow of solvent over the drug column is generally downward to the exit orifice, impelled by force of gravity as well as the weight of the column liquid.
- Preparation of crude drugs for percolation
  - Crude drug is Powdered
  - Moistened to encourage swelling
  - Period of maceration
  - Packed in the percolator
  - Percolation and collection of percolate

- Advantages include
  - Suitable for thermolabile compounds
  - No need of filtration
  - Problem of saturation reduced
- Disadvantages
  - Not suitable for unorganized drugs.
  - Time taking process ⇒ possibly of hydrolysis of compounds
  - Usually done at room temperature

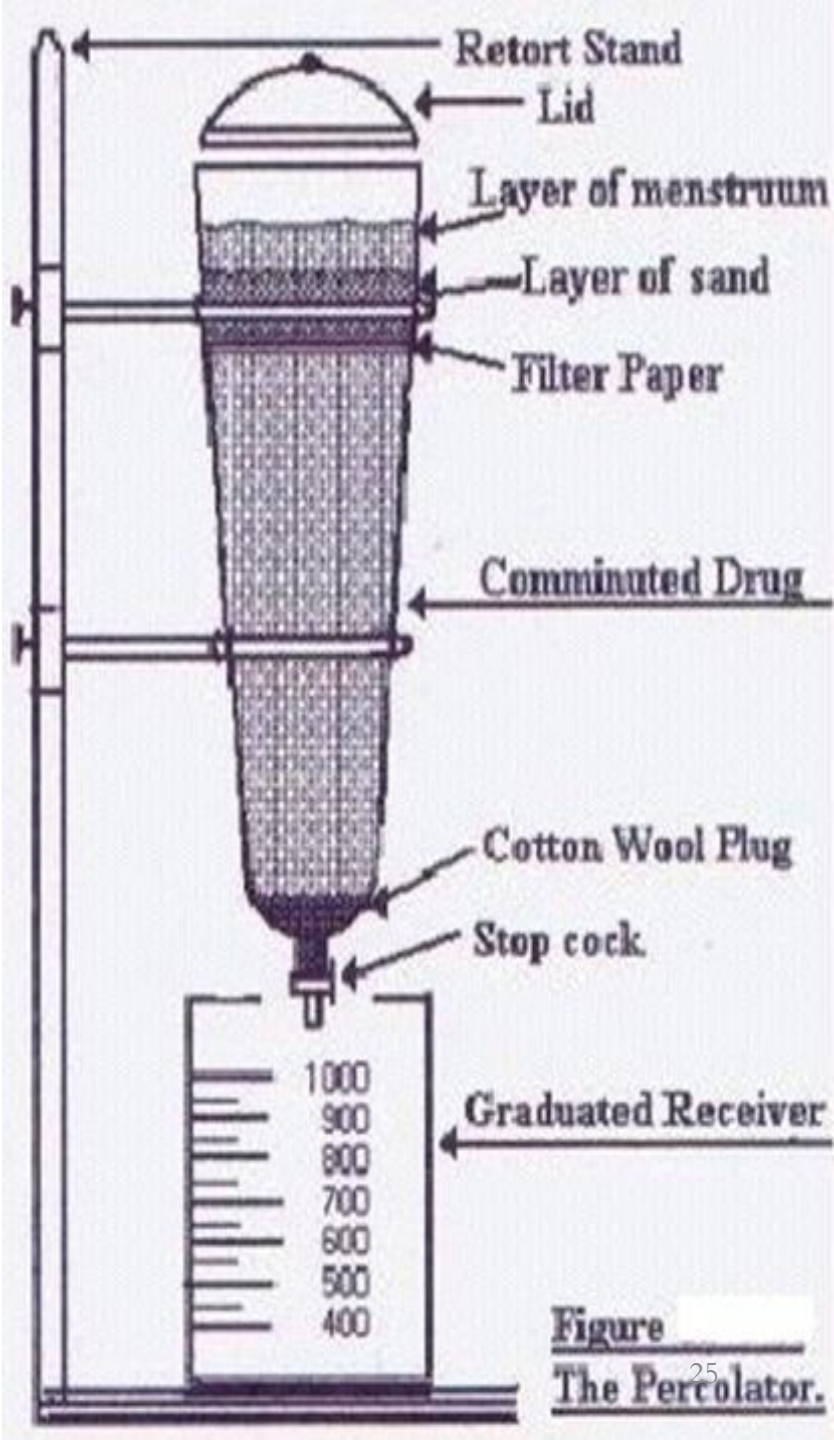


Figure
The Percolator.

Continuous hot percolation (Soxhlet)

- A Soxhlet extractor is a piece of laboratory apparatus
  - invented in 1879 by Franz von Soxhlet.
  - It was originally designed for the extraction of a lipid from a solid material.
  - A Soxhlet extractor is not limited to the extraction of lipids.
  - A Soxhlet extraction is only required where the desired compound(s) has only a limited solubility in a solvent, and the impurity is insoluble in that solvent.

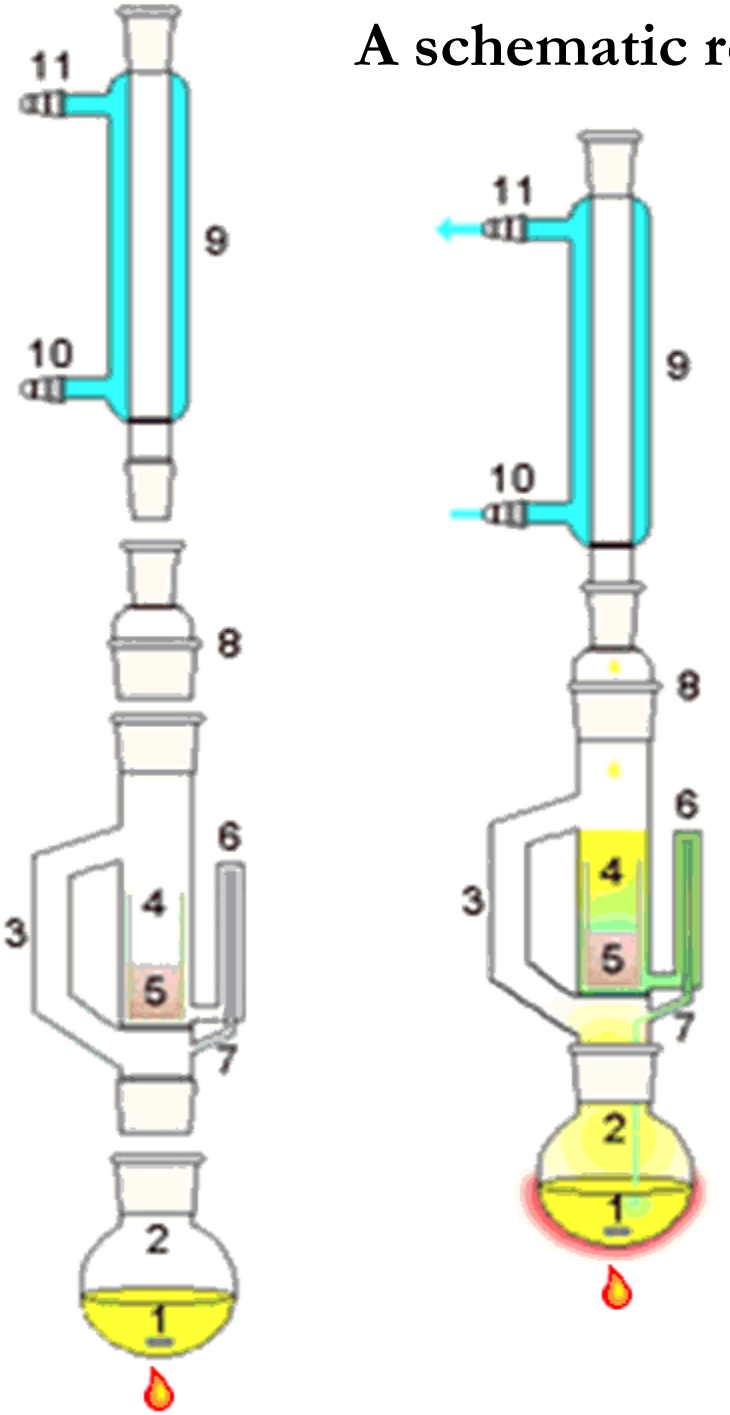
Components of soxhlet extraction set up

- Heat source
- Flask containing solvent
- Extraction chamber
- Thimble (made of filter paper)
- Water jacketed Condenser

Procedure:

- Powdered plant material is
  - packed in the thimble and
  - he mouth of the thimble covered by piece of cotton.
  - The thimble is made from thick filter paper
- The packed thimble is put in the extraction chamber and enough amount of volatile solvent is taken in the round bottom flask.
- The extraction chamber is placed on the round bottom flask containing the solvent.
- A water jacketed condenser is then connected to the upper end of the extractor.

A schematic representation of a Soxhlet extractor



1. Stirrer or pieces of porcelain
2. Round bottom flask with solvent
3. Vapour path(side tube)
4. Thimble
5. Powdered crude drug
6. Siphon top
7. Siphon exit
8. Expansion adapter
9. Condenser
10. Cooling water inlet
11. Cooling water outlet

- After all the components are fitted appropriately, water is let to flow through the condenser and the heater is switched on.
  - The condenser ensures that any solvent vapour cools and drips back into the chamber housing the solid material.
  - Thus, the heated solvent starts to evaporate and the vapor passes through the large arm of the extraction chamber and reaches the condenser where it will condense.

- The condensed solvent drips on to the thimble and starts to accumulate in the extraction chamber, extracting the plant material in the process.
- The chamber containing the solid material slowly fills with warm solvent.
- Some of the desired compound will then dissolve in the warm solvent.
- When the Soxhlet chamber is almost full, the chamber is automatically emptied by a siphon side tube, with the solvent running back down to the round bottom flask.

- This cycle may be allowed to repeat many times, over hours or days.
- During each cycle, a portion of the non-volatile compound dissolves in the solvent.
- After many cycles the desired compound is concentrated in the distillation flask.
- After extraction the solvent is removed, typically by means of a rotary evaporator, yielding the extracted compound.
- The non-soluble portion of the extracted solid remains in the thimble, and is usually discarded.

Continuous hot percolation(Soxhlet)

- **Advantages**
  - Saves solvent
  - Efficient
  - Saturation is not a problem i.e. instead of many portions of warm solvent being passed through the sample, just one batch of solvent is recycled.
- **Disadvantages**
  - Not suitable for thermolabile components
  - Expensive

Infusion

- This extraction is by taking hot solvent and the contact time between the plant material and the solvent is relatively short.
- This method is used for those drugs which are soft in nature so that water may penetrate easily to the tissues and the active constituents are water soluble.
- Infusions must be freshly prepared and consumed within 24 hrs of its preparation.
- An infusion containing about 20-25% alcohol can be stored for sufficient long time. Alcohol is used as a menstruum or as a preservative.

  Ex. Infusion of senna and quassia.

Digestion

- This extraction is carriedout by applying gentle heat to the substances being extracted.
- This method is applicable only to those drugs where moderately elevated temperature is not objectionable and the solvent action of the menstruum is increased by gentle heat.

Decoction

- This extraction is by boiling
- The plant material is boiled in the solvent for about 10-15minutes. After boiling the liquid is cooled and filtered., more water is passed through the marc to produce the required volume. A freshly prepared decoction should only be dispensed and the same must be consumed within 24hrs.



  - Advantage:
    - Simple
    - Quick method
  - Disadvantage:
    - Thermolabile compounds will be destroyed
    - Wastage of solvent

Refluxing

Refluxing means boiling a solvent in a vessel attached to a condenser so that the vapours continuously condense for reboiling.

Procedure

- Powdered plant material is placed in a flask
- Sufficient amount of the menstruum is added
- The flask is connected to a condenser, such that any vapours entering the condenser are cooled back to liquid, and fall back into the flask.
- The flask is then heated vigorously for the course of extraction (usually 48hrs).
- Cool the content of the container under running water and then filter.Concentrate the filtrate

Refluxing

- Advantage
  - it can be left for a long period of time
    - without the need to add more solvent or
    - fear of the flask content boiling dry as any vapour is immediately condensed in the condenser.
  - The solvent will always boil at a certain temperature,
  - The extraction will proceed at a constant temperature.
- Disadvantage
  - It is not suitable for thermolabile plant constituents

Distillation methods

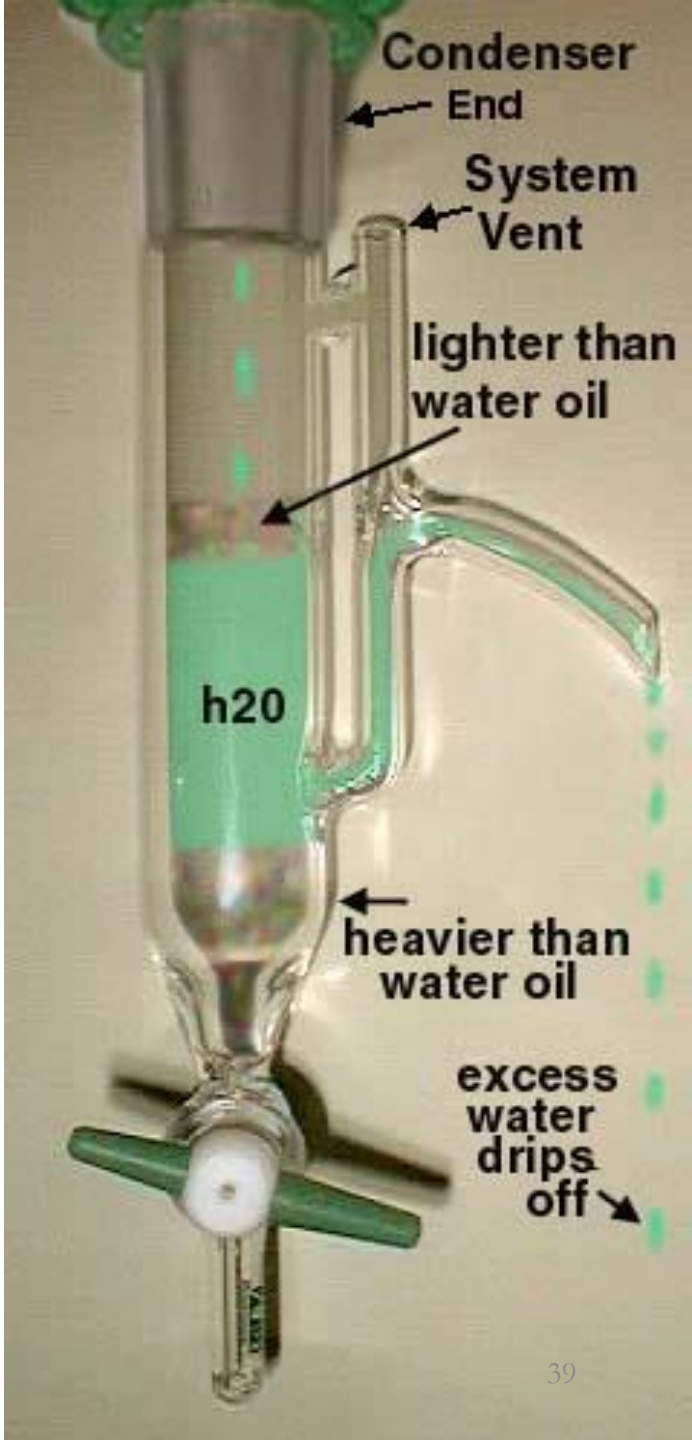
- Water distillation
- Water and steam distillation
- Direct steam distillation

Water distillation

- Water or hydro distillation is one of the oldest and easiest methods being used for the extraction of essential oils.
- In this method the plant material is fully dipped in the water.
- Plant material is introduced with water into the distilling chamber and subjected to heat until both volatile matter and water, are condensed.

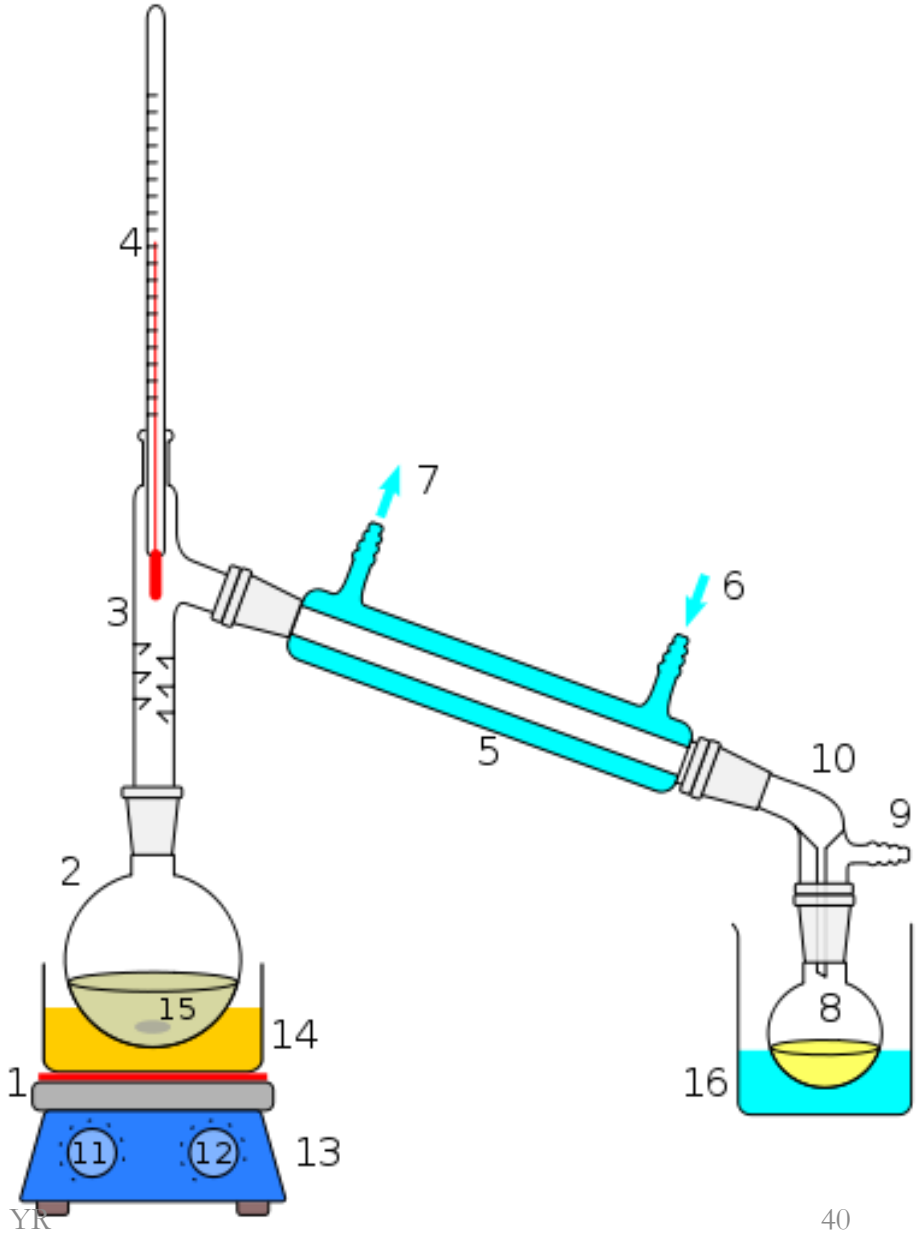
–e.g. turpentine oil

- Applied to plant material that is dried and not subject to injury by boiling.



Water distillation

1. Heat source
2. Still pot
3. Still head
4. Thermometer
5. Condenser
6. Cooling water in
7. Cooling water out
8. Distillate/receiving flask
9. Vacuum/gas inlet
10. Still receiver
11. Heat control
12. Stirrer speed control
13. Stirrer/heat plate
14. Heating (Oil/sand) bath
15. Stirrer bar/anti-bumping granules
16. Cooling bath.



Disadvantages:

- As the plant material near the bottom walls of the still comes in direct contact with the fire from furnace, there is a likelihood of its getting charred and thus imparting an objectionable odour (burning note) to the essential oil.
- Prolong action of hot water can cause hydrolysis of some constituents of the essential oils such as ester, etc.
- The process is slow and the distillation time is much longer thereby consuming more firewood / fuel.

Water and steam distillation:

- To remove the drawbacks of water distillation the design has been improved where a perforated grid / net is introduced into the still just above the bottom.
- This method is employed for plant materials, either dried or fresh that their volatile oil may injured by boiling e.g. Cinnamon and Clove.
- If dried, plant material is grounded, or chopped if fresh and is covered with water and steam is then passed through the macerated mixture to avoid direct boiling.
- Steam is generated else where and piped in to the flask containing the drug
  - Oil is separated from the aqueous layer

- Plant material mixed with water in mother flask and also injected with steam.
- Water and steam must be in contact with surface of plant material.
- As the hot steam passes through the mixture it condenses, releasing heat.
- Used to extract oils from wood, bark, seeds and dried buds.

Direct steam distillation:

- The steam distillation process for the extraction of essential oils from plant material consists of 3 basic parts - the retort, the condenser, and the separator.

The Retort or Still

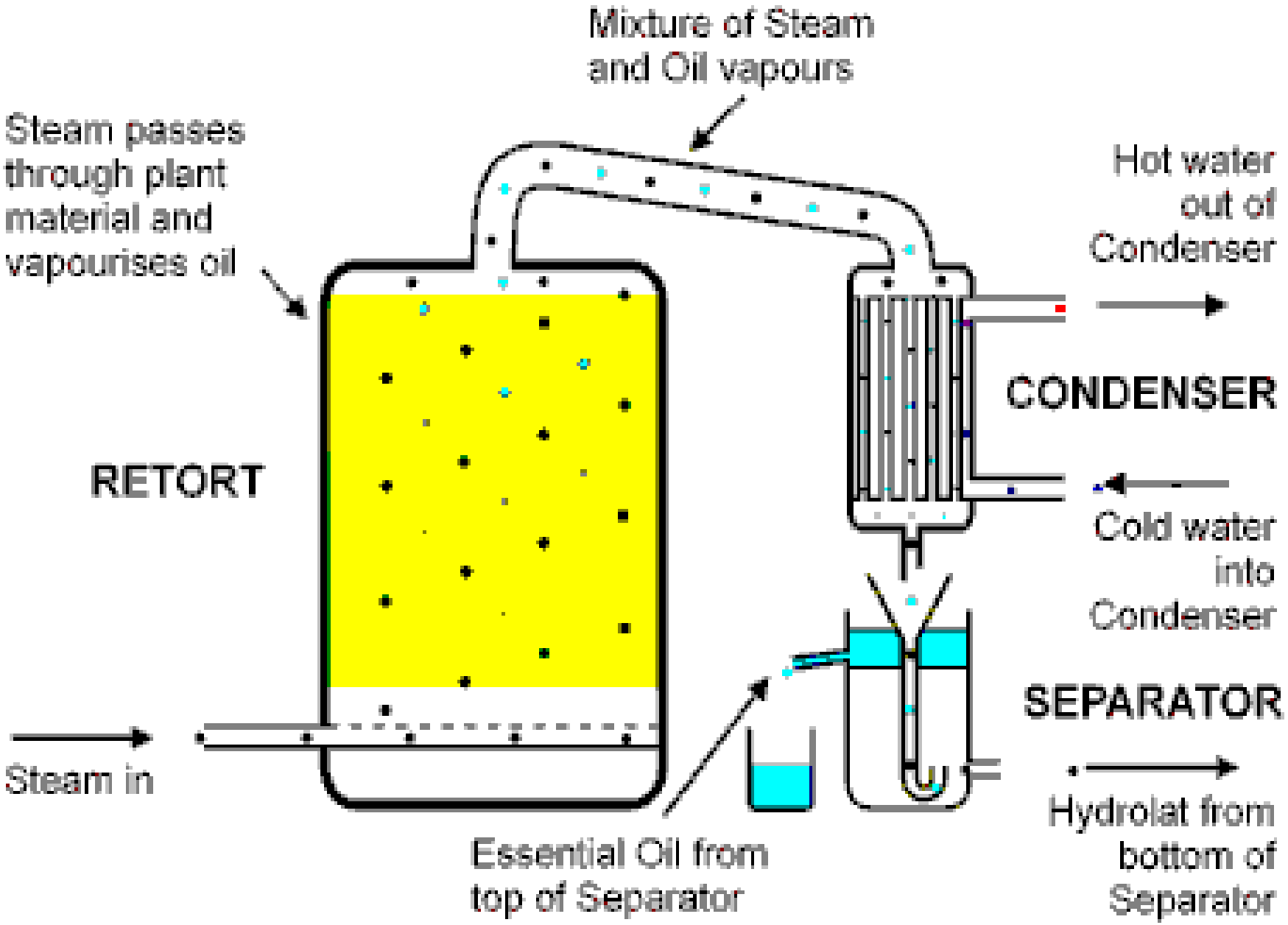
- This contains the plant material containing the volatile essential oil.
- High pressure steam is passed in at the base into a void beneath a perforated grid which supports the charge of green herb.
- The steam passes through the plant material heating and saturating it with water.
- The resulting vapour, a mixture of steam and essential oil vapours, passes out at the top and is conveyed to the condenser.

The Condenser

- The vapours from the retort are cooled by passing down a long spiral or bundle of tubing or plates cooled by being immersed in a flow of water.
- The condensate runs by gravity into a container in which the oil and water separate by virtue of their differing specific gravities in the separator.

The Separator

- The mixture of condensed oil and water runs into the separator where the lighter insoluble oil floats on the surface, accumulates slowly and is drawn off periodically. This is the **essential oil.**


Mixture of Steam and Oil vapours
Steam passes through plant material and vapourises oil
Hot water out of Condenser
CONDENSER
RETORT
Cold water into Condenser
SEPARATOR
Steam in
Hydrolat from bottom of Separator
Essential Oil from top of Separator

Expression

- Expression as a method of frangrance extraction where raw materials are pressed, squeezed or compressed and the oils are collected.
- In contemporary times, the only fragrant oils obtained using this method are the peels of fruits in the citrus family.
- This is due to the large quantity of oil is present in the peels of these fruits as to make this extraction method economically feasible.
- Citrus peel oils are expressed mechanically, or *cold-pressed*.

Enfleurage

- Enfleurage is a two-step process during which the odour of aromatic materials is absorbed into wax or fat, then extracted with alcohol.
- Extraction by enfleurage was commonly used when distillation was not possible because some fragrant compounds denature through high heat.
- This technique is not commonly used in modern industry, due to both its prohibitive cost and the existence of more efficient and effective extraction methods.